

HANNAH NAGPALL

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SUMMARY

Honors Computer Science student (GPA 3.95, 4.0 Spring 2026) at Texas A&M University–Kingsville with experience in software engineering, machine learning, and AI research. Developed production backend systems using Java/Spring Boot and contributed to multiple research projects spanning medical AI, privacy auditing, and wireless communications. Author and co-author of several IEEE and journal manuscripts with interests in trustworthy AI, healthcare AI, and intelligent systems.

EDUCATION

B.S. Computer Science (Honors) · Texas A&M University–Kingsville

Aug 2025 – Present | GPA: 3.95 (4.0 current semester)

Honors College Scholar · Presidential List Spring 2026 · Dean's List Fall 2025

B.S. Computer Science · Alabama A&M University · NCAA D1 Scholar-Athlete (Tennis, Full Scholarship)

Aug 2023 – May 2025

· NRF Ray Greenly Scholarship · Presidential Silver Medallion Spring 2025 · Presidential Freshman Award

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS **Backend:** Spring Boot, REST API **Databases:** MongoDB, PostgreSQL, MySQL

ML / Data: scikit-learn, pandas, NumPy, SciPy, matplotlib, NNLS, SHAP, LIME, XGBoost **Tools:** Git, GitHub, Flask, Ollama, AWS

EXPERIENCE

Research Collaborator- Behavioral Robustness · TAMUK Dept. of Computer Science

May- Present

- Investigating the robustness and physiological consistency of temporal health prediction models under real clinical interventions.

Research Assistant — Raman Spectroscopy · TAMUK Dept. of Chemical & Natural Gas Engineering

Nov 2025 – Present

- Built a spectral analysis pipeline for multi-component hydrocarbon mixtures implementing peak detection, Lorentzian fitting, and NNLS-based composition ratio estimation across 500–3500 cm⁻¹. Applied preprocessing (baseline correction, normalization, smoothing);

Backend Development Intern · GSPANN Technologies, Inc.

May – Aug 2025

- Developed backend components for an e-commerce platform using Java, Spring Boot, and MongoDB, including user authentication, product catalogue, and order handling.

AI in Retail Technology Extern · The Home Depot

May – Jul 2025

- Benchmarked 3 AI risk frameworks for retail supply chain; delivered Risk-Landscape Summary adopted as reference material across the extern cohort.

RESEARCH & PUBLICATIONS

AI Physiological Alignment- Behavioural Robustness · *Ongoing*

- Investigating behavioral robustness and physiological consistency in temporal health prediction models under simulated interventions.

Raman Fuel Analysis System · *Abstract accepted at AIChE*

- Spectral analysis pipeline for multi-component hydrocarbon mixtures with Lorentzian peak fitting, NNLS composition estimation, and normalization workflows.

Memory That Remembers Wrong: Confirmation-Loop Bias in LLM Health Agents · *AI4S Conference (Under Review)*

- Developed a counter-evidence framework to mitigate confirmation-loop bias in LLM health agents using MIMIC-IV clinical data.

The Faithfulness Paradox in Hate Speech Detection · *AI4S Conference (Under Review)*

- Comparative evaluation of explanation faithfulness and energy efficiency across linear and neural hate-speech classifiers.

Energy-Efficient Fluid Antenna Arrays for AirComp · *IEEE WCL (Under Review)* · [zenodo](#)

- Proposed a joint optimization framework for port selection and power control in fluid antenna-enabled over-the-air computation systems.

Detecting Mixed-Authorship Academic Writing: A Leakage-Controlled Framework · *SN Computer Science (Under review)* · [zenodo](#)

- Three-class AI/Hybrid/Human detection system with 51-feature pipeline (stylometric, semantic, citation); leakage-controlled evaluation; macro-F1 0.611. Honors project, TAMUK.

PrivAudit: Interpretable Membership Inference Risk Auditing · *SN Computer Science (Under Review)* · [zenodo](#)

- Lightweight CPRS framework for ML privacy auditing across 12 tabular benchmarks; r=0.846 correlation with empirical MIA balanced accuracy. Honors project, TAMUK

Decoupling Transient Instability & Steady-State Persistence in Nonlinear Multi-Agent Networks · *IEEE TCNS (Under Review)* · [zenodo](#)

- Nonlinear influence-recovery simulation framework for stability dynamics across graph topologies.

PROJECTS

AI Health Companion · React · TypeScript · JavaScript · Python · CSS · Rest API

2026-Present

- Developing an intelligent health companion platform for personalized health tracking, adaptive wellness planning, and AI-assisted preventive care.

TOEFL Practice Platform · HTML · CSS · JavaScript · Flask/Python

2026-Present

- Built an online TOEFL preparation website offering section-wise practice tests for Reading, Listening, Speaking, and Writing.

F1 Race Predictor · Python · scikit-learn · pandas

2025

- ML model predicting F1 race outcomes using historical data (lap times, qualifying, weather, driver stats); feature engineering and model selection pipeline.

EduQuest Advisors Platform · HTML · CSS · GitHub Pages

2025

- Founded and shipped live web platform for India's first à-la-carte college tennis placement service targeting AITA-ranked players.

AWARDS & RECOGNITION

Project Q2 Global Research Hackathon · Honors College Scholar · NRF Ray Greenly Scholarship · Presidential Freshman Award · Presidential Silver Medallion Spring 2025 · Dean's List Fall 2025 · Presidential List Spring 2026 · NCAA D1 Full Athletic Scholarship · HBCU National Champion

LEADERSHIP

Community Volunteer · Samidha Welfare Foundation — Ongoing. Supported on-ground coordination and event execution across service drives for underprivileged communities · VC Research & Pitch Desk Extern